Within-Region Power Analysis Plan

Focal (local) oscillatory activity across sensory preconditioning: Precon 1 vs Precon 4

Draft for discussion | October 29, 2026 | Companion to the coherence analysis plan; based on Dickson et al. (2022), eNeuro 9(2): ENEURO.0414-21.2022

1. Overview

Coherence measures coupling **between** regions. To contrast focal change with distributed (network) change, this plan adds a measure of local oscillatory activity: **power within each region**, across frequency bins within the θ and γ bands. Power is compared between baseline, cue 1 and cue 2 on Precon 1, and again on Precon 4.

Design at a glance

Question	Does oscillatory power within a region change from early (Precon 1) to late (Precon 4) preconditioning, and does the cue-related change differ between sessions?
Animals	n = 8, analyzed as a single cohort (no between-subject factor)
Regions	RSC, dHC, ACC, OFC, PER (POR excluded because it was not reached); recorded bilaterally, averaged across hemispheres
Session factor	Precon 1 vs Precon 4
Period factor	Baseline vs cue 1 vs cue 2
Frequency bands	θ and γ, each analyzed separately, as a band average and across frequency bins
Pooling	Cue 1 pools cues A and C. Cue 2 pools cues B and D (counterbalanced). Cue pairs are pooled within each session.

Notation: n (Factor) × m (Factor) lists each factor with its number of levels. DV = dependent variable. f = frequency bin (bins are specific to each band).

2. Statistical model

Analyses use a **generalized estimating equation (GEE)**, as in Dickson et al., with animal ID as the subject (cluster).

Subject variable	Animal ID
Within-subject variables	Session, Period, Region, Frequency bin
Working correlation	Exchangeable
Distribution and link	Log or dB power is roughly normal: identity link, normal distribution. Alternatively, gamma with a log link on raw power. Check the residuals.
Estimator	Robust (sandwich) standard errors, ideally with a small-sample correction, because n = 8 is small for GEE.
Alternative	Linear mixed model with a random intercept for animal and Kenward-Roger degrees of freedom. If both approaches agree, report the second as a robustness check.

3. Power analyses

Each analysis below is run **separately for θ and for γ**.

E1 Band-averaged power

Design	2 (Session: Precon 1 vs Precon 4) × 3 (Period: baseline vs cue 1 vs cue 2) × 5 (Region: RSC, dHC, ACC, OFC, PER)
DV	Band-aveer (dB)
Tests	Session × Period (key effect), Session × Period × Region, Period
Follow-up	Session × Period within each region, FDR across the 5 regions. Planned Period contrasts: cue 1 vs baseline, cue 2 vs baseline, cue 2 vs cue 1.

E2 Power across frequencies

Design	2 (Session: Precon 1 vs Precon 4) × 3 (Period) × 5 (Region) × f (Frequency bin)
DV	Log power (dB) at each frequency bin
Tests	Session × Period × Frequency, Session × Period × Region × Frequency
Fallback	2 (Session) × 3 (Period) × f (Frequency bin), run once for each of the 5 regions, FDR across regions (use if the full model is unstable at n = 8)

E3 Oscillation vs 1/f background (optional)

Design	2 (Session) × 3 (Period), run once for each region and each parameter
DV	Peak frequency, peak amplitude and aperiodic slope, from spectral parameterization (specparam / FOOOF)
Purpose	Separates a change in the oscillation itself from a broadband (1/f) shift, which raw power across bins cannot do.

> **Session × Period is the key effect.** Baseline is recorded in the same session as the cues, so slow signal drift across days (impedance, gain) cancels in this interaction. The Session main effect does not have that protection, so do not interpret it on its own.

Method notes

- **Baseline:** a 10-s window before cue 1 onset, matched in duration to the cues and clear of any previous cue.
- **Cues:** cue 1 pools A and C, and cue 2 pools B and D (counterbalanced).
- **Cue-onset response:** trim the first 1-2 s or subtract the trial-averaged evoked response, since it inflates low-frequency (θ) power.
- **Matched pipeline:** same windows, hemisphere averaging and frequency bins as the coherence analyses. Ideally also compute coherence in the same three periods.
- **Model:** log or dB power is roughly normal, so use a GEE with an identity link and normal distribution (or gamma with a log link on raw power), clustered on animal.
- **Order confound:** cue 2 vs cue 1 is confounded with time in the trial. The change across sessions is what protects the interpretation.

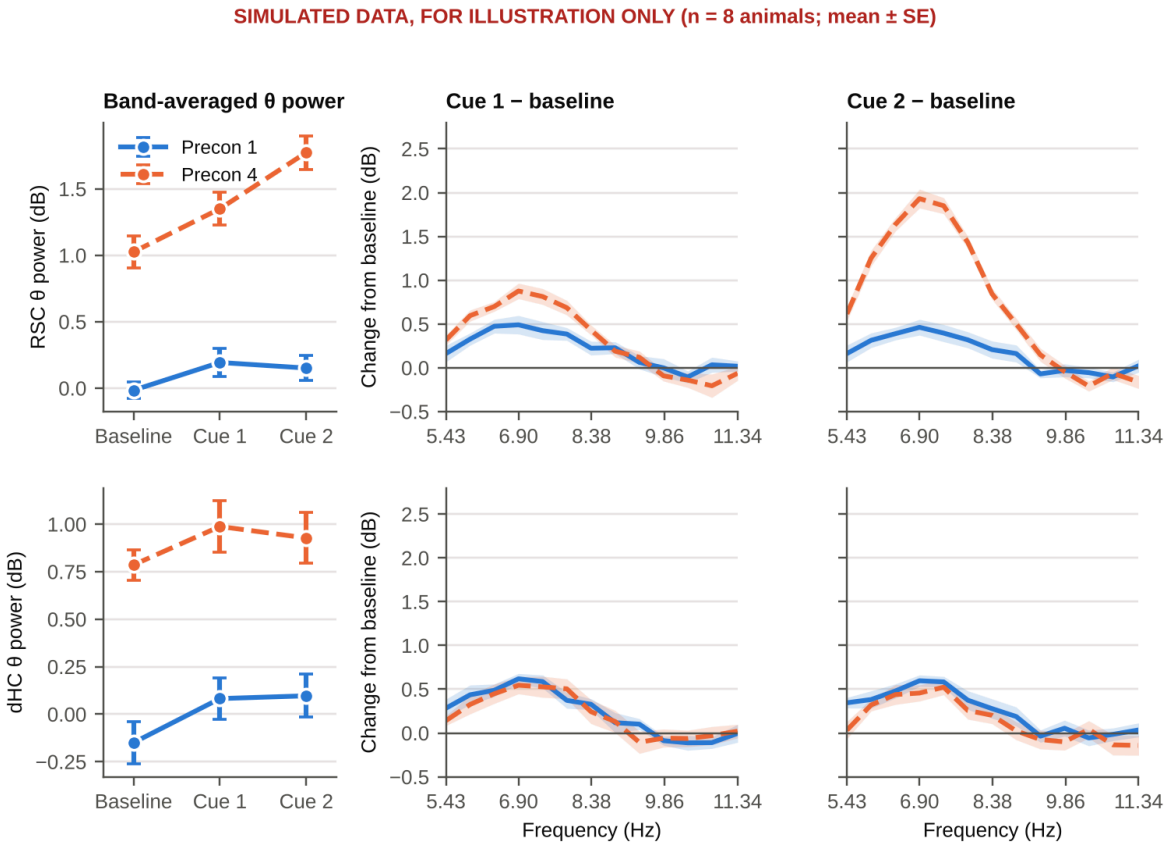
Reading power together with coherence

Power (E1-E2)	Coherence analyses	Interpretation
Changes	**No change**	Local (focal) change in that region
No change	**Changes**	Change in inter-regional coupling
Changes	**Changes**	Local and network change together
No change	**No change**	No detectable change

"Focal" is an operational label here, not proof. A region's power can rise because of network input, so report these as local oscillatory change vs inter-regional coupling.

Example output (simulated data)

The figure below uses **simulated data** to show the intended format. Rows are regions and columns are band-averaged θ power (left) and the change from baseline across frequency for cue 1 and cue 2 (middle, right). Precon 1 is solid and Precon 4 is dashed. In this example, RSC shows a larger cue-related increase on Precon 4 (a Session × Period effect, strongest for cue 2 near 7 Hz), while dHC does not change. Both regions have a higher baseline on Precon 4, which stands in for day-to-day drift and does not affect the interaction.



Simulated data for illustration only (n = 8, mean ± SE). Not real results.